SCI-POINT

Engineering Conformance Specification

Scientific contract v0.1 Document revision r0.2

Conditional generic Stage B contract

Conformance status. This specification is normative but conditional. No numerical POINT fit conforms until the exact compatibility method, exact numerical parent, and route binding are owner-approved. Formal errors and full-map-RMS diagnostics have separate gates. No implementation has been assessed against this specification.

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Conformance rule

An implementation or product is conforming only for the requirements whose complete dependencies are available and whose exact identities and states are recorded. Typed unavailability is conforming when a dependency is absent and the contract requires fail-closed behavior. Fabricating a default, zero, alias, route substitution, association, covariance, response, bias, or use decision is nonconforming.

The shared common core is the normative scientific source. The rationale explains that source. This ECS supplies prospective engineering evidence procedure and authors no independent science. A change to any common-core byte requires a new binding and document generation.

Contents

1 Normative notation and state axes

The words **shall**, **shall not**, **required**, and **prohibited** are normative. "Available" means that every dependency for the stated quantity is bound and valid. "Unavailable" means that the quantity or claim has not been established; it never means numerical zero. "Failed" means that an authorized computation was attempted but did not return a conforming result.

Symbol or term	Meaning
m_q	Exact parent signal at admitted row or pixel q.
$\boldsymbol{u}_q$	Physical coordinate supplied by the exact AltAz tangent-boundary transform for parent coordinate p_q.
A	Fitted amplitude in the exact parent product's unit, calibration, and response.
$\boldsymbol{\mu}$	Continuous fitted centroid in the declared ordered AltAz tangent basis.
$\boldsymbol{\Sigma}_g$	Symmetric positive-definite invariant Gaussian shape matrix.
$\boldsymbol{\theta}$	The six scientific roles: amplitude, two centroid components, two principal-width roles, and orientation.
$\mathcal{P}$	The complete data-dependent POINT procedure used to define a full-procedure response.
P_μ	Selector that extracts the two fitted-centroid components from the parameter vector.
W_{fit}	Exact fit-weight role, distinct from parent covariance, inverse covariance, reliability weight, and uncertainty products.
$\boldsymbol{\Delta}_{\mathrm{POINT}}$	Measured source displacement, fitted centroid minus authoritative expected source position, in arcseconds in the declared basis.
parent	One exact observation-local, per-array MAP, JINC, FLT-FIXED, or FLT-MATCHED signal product.
terminal FRUIT ancestry	Lineage attached to one of the four parent families; it is not a fifth parent family.
named use	One owner-defined consumer purpose evaluated without changing the immutable POINT result.

1.1 Three independent state systems

Producer lifecycle, component identifiability, and named-use evaluation shall remain independent.

Producer lifecycle `not_requested`, `requested`, `effective`, `unavailable`, `resolved`, `applied`, `fit_realized`, `complete_publication_candidate`, `publication_decided`, `published`, `failed`, `not_produced`, or `superseded`.

Component identifiability Exactly one of `available_identifiable`, `available_bound_censored`, `weakly_identified`, `undefined_by_model_symmetry`, `unavailable`, or `failed` for every required parameter role.

Named-use evaluation Four fields, kept separate:
request: `requested` or `not_requested`;
applicability: `applicable`, `inapplicable`, or `applicability_unknown`;
eligibility: `eligible`, `ineligible`, or `decision_unavailable`;
realization: `realized`, `incomplete`, `failed`, or `not_produced`.

After the four named-use fields are evaluated, the owning consumer may prescribe an action such as `diagnostic_display_only`. That token is an action, not an eligibility value, producer state, or universal quality flag. The token `diagnostic_only` is prohibited as a producer state or SCI-VAL eligibility value. A generic product-availability field may use `unavailable`; that token shall not be substituted into a SCI-VAL realization field. Likewise, `decision_unavailable` is an eligibility value only and shall not appear as applicability.

1.2 Exact SCI-VAL disposition function

For one named use, the evaluator shall apply the following precedence and shall emit all four axes. An already realized producer fact is never erased by the named-use disposition.

Controlling condition	Request	Applicability	Eligibility	Realization
Use not requested	`not_requested`	retained fact or `applicability_unknown`	`decision_unavailable`	`not_produced`
Missing profile or source binding	`requested`	`applicability_unknown`	`decision_unavailable`	`not_produced`
Structural profile/binding conflict	`requested`	`applicability_unknown`	`decision_unavailable`	`failed`
Use factually inapplicable	`requested`	`inapplicable`	`ineligible`	`not_produced`
Applicable, artifact incomplete	`requested`	`applicable`	`decision_unavailable`	`incomplete`
Applicable, artifact failed	`requested`	`applicable`	`decision_unavailable`	`failed`
Applicable, artifact not produced	`requested`	`applicable`	`decision_unavailable`	`not_produced`
Applicable and policy-evaluable	`requested`	`applicable`	`eligible` or `ineligible`	`realized`, `incomplete`, `failed`, or `not_produced`, as observed

Every row also carries the exact controlling cause, profile identity, source bindings, evaluation identity, lifecycle, provenance, and any prescribed post-evaluation action. Missing-source means a missing source authority or source binding required by that exact use; it does not delete an otherwise conforming unassociated fit component.

2 Scientific operation and ownership

SCI-POINT fits one known, isolated, bright Pointing source on one exact observation-local per-array map product. Each requested observation-array-parent-method fit is scientifically atomic. The primary scientific result, when all dependencies are available, is a two-component source displacement in a declared AltAz tangent basis. The complete fit also has amplitude, centroid, two width roles, orientation, support, constraints, method identity, parent identity, uncertainty state, association state, and use-specific evaluation facts.

A POINT fit atom is complete when every required role is present with one exact value-or-status state, identity, cause, and provenance. Atomicity does not require every role to contain a finite numerical value. Exact circularity can therefore carry a complete orientation role with `undefined_by_model_symmetry`; one width may be weakly identified while the centroid remains available; and formal errors, response, bias, or source association may be unavailable without deleting conforming numerical fit facts. The POINT-owned fit-completeness profile determines eligibility only for its named complete-fit use and never erases available producer facts.

SCI-POINT is a producer of per-array measurements. It does not detect blind sources, deblend, catalog, make or filter maps, run FRUIT recurrence, calibrate, infer per-detector beams, perform OOF inference, aggregate arrays, form a telescope correction, select a correction, or apply one.

2.1 Producer-transformer-consumer boundary

The exact parent producer owns its estimand, signal unit, calibration, normalization, WCS, grid, frame, tangent relation, support, validity, missing/non-finite policy, response, covariance or uncertainty state, null-space, lifecycle, and provenance. POINT binds and consumes those facts without redefining them.

The source-reference producer owns the expected source position and its identity, convention, time or epoch, frame, atmospheric and aberration treatment, morphology or reference origin, uncertainty, validity, lifecycle, and provenance. AST owns the exact AltAz tangent-basis realization. A named pointing-support producer owns cross-array membership, aggregation, weights, dependence, partial-set policy, measurement-to-correction sign, telescope user/paddle-offset composition, selection, native support, and correction record publication. AST owns conforming application.

SCI-BEAM owns per-detector Beammap fits, detector PSF, sensitivity, APT, and beam authority. CAL or TolProj owns photometric-transfer admission and reference-flux authority. A named telescope/observing-condition QC process owns its reference populations, thresholds, comparisons, aggregation, trend analysis, actions, and causal interpretation. NOI owns separately authorized empirical uncertainty. SCI-VAL registers and evaluates exact profiles but authors none of their scientific policies.

2.2 Eligible parent families are not equivalent

Family	Exact POINT signal role	Required route distinction
MAP	`MAP-SIGNAL/OBSERVATION-LEVEL-NORMALIZED@1`	POINT-facing boundary-role identity for an exact observation-level normalized upstream signal; never a coadd.
JINC	`JINC-SIGNAL/NORMALIZED-JINC-MAP@1`	POINT-facing boundary-role identity for an exact signed and normalized upstream signal; N, C, Q, coefficient-squared accounting, response, covariance, and state records are companions, not alternate signal parents.
FLT-FIXED	`FLT-FIXED-SIGNAL/TRANSFORMED-MAP@1`	POINT-facing boundary-role identity for an exact upstream transformed signal with its MAP/JINC parent, filter operator, normalization, response, edge/support, and covariance state.
FLT-MATCHED	`FLT-MATCHED-SIGNAL/MATCHED-TEMPLATE-AMPLITUDE-FIELD@1`	POINT-facing boundary-role identity for an exact upstream matched-template amplitude field with template, noise/covariance selector, normalization, phase/origin, response, and complete support.

POINT shall not choose, substitute, equate, or fall back among routes. The route-specific chain from source morphology through parent response to the expected processed-map profile and Gaussian compatibility shall be bound for each instance. Every numerical route currently has state `unavailable_pending_method_and_route_binding`. Eligibility of a family does not establish an exact numerical parent, Gaussian adequacy, response fidelity, zero centroid bias, or numerical availability.

These four tokens are POINT-facing boundary-role identities, not assertions that an upstream package publishes the same literal token. Every approved numerical boundary instance shall map the POINT-facing role to one exact upstream package product role, contract version, source digest, owner approval, compatibility state, and generation. A POINT-side label alone proves neither upstream existence nor upstream naming.

A terminal FRUIT product retains its exact MAP, JINC, FLT-FIXED, or FLT-MATCHED identity and adds the complete method, terminal iteration, generation, recurrence, restart, response, support, covariance, calibration, null/additive-reference, lifecycle, and provenance ancestry. The record shall assert that the product is terminal. Intermediate iterations and coadds are outside base v0.1.

2.3 Product roles and claim ceilings

Role	Meaning and ceiling
`POINT-FIT-RESULT`	Conditional numerical six-role fit with lifecycle, component states, support, method, and parent; not association, a catalog, an intrinsic beam, or a correction.
`POINT-FIT-AMPLITUDE-COMPONENT`	Numerical A role of an otherwise conforming fit. It may exist with association unavailable or failed, but retains that state and is not attributed to the intended source.
`POINT-DISPLACEMENT-MEASUREMENT`	Source-associated fitted centroid minus expected position in the exact AltAz basis; not an absolute spherical position or correction vector.
`POINT-SOURCE-ASSOCIATED-AMPLITUDE-DIAGNOSTIC`	Fit-amplitude component with `POINT-SOURCE-ASSOCIATION-STATE=established`; the only amplitude role that may be a candidate input to CAL/TolProj admission. It is not universal flux or pre-established photometric eligibility.
`POINT-DYNAMIC-RANGE-DIAGNOSTIC/FITTED-AMPLITUDE-OVER-FULL-MAP-RMS@1`	Independently requested derived product under `POINT-FULL-MAP-RMS-METHOD v0.1`; descriptive unit-one dynamic range only.
`POINT-FORMAL-AMPLITUDE-STANDARDIZATION@1`	Independently requested derived product under `POINT-FORMAL-ERROR-METHOD v0.1`; formal unit-one standardization only.

Role	Meaning and ceiling
`POINT-EFFECTIVE-SHAPE-DIAGNOSTIC`	Required width roles and angle under the processed-map response; not an intrinsic telescope beam, detector PSF, SCI-BEAM product, or unique cause.
`POINT-FIT-QUALITY-CONTROL-METRICS`	Centroid, amplitude, shape, support, constraints, uncertainty, residual/response state, and fit state for an owner-governed QC use; not an automatic threshold or action.
`POINT-SOURCE-ASSOCIATION-STATE`	`established`, `unavailable`, or `failed`, with method, domain, cause, lifecycle, and provenance; never implied by a peak or finite fit.
`POINT-FIXED-BRANCH-RESPONSE-STATE`	Response conditional on one exact branch, support, weight state, and active constraints; not full-procedure response.
`POINT-FULL-PROCEDURE-RESPONSE-STATE`	Response obtained by rerunning every data-dependent branch under perturbation; not observational accuracy.
`POINT-OBSERVATIONAL-BIAS-ACCURACY-STATE`	Separate empirical authority; unavailable bias is not zero bias.
`POINT-UNCERTAINTY-STATE`	Separately typed formal, coordinate, reference, response, empirical, NOI, calibration, dependence, and correction uncertainty; not an implied total.

2.4 Applicability facts

The four facts `known`, `isolated`, `bright`, and `approximately_centered` each carry value/state, exact authority or method, domain, cause, validity, lifecycle, and provenance. `known` requires a valid source identity and expected-position record. `isolated` requires authority over confusing sources or structure in the search and fit domains. `bright` is a compatibility-method class, not significance or either diagnostic ratio. `approximately_centered` is an exact relation among expected position, search domain, fit domain, parent support, and map boundary. Tests for the latter three remain unavailable pending method authority. Labels, filenames, defaults, map finiteness, fitted amplitude, or ratios shall not instantiate them.

2.5 Draft named-use profile identities

The following collision-free mechanics are drafts for final owner review, not registered profiles:

- `VAL-PROFILE/SCI-POINT/FIT-COMPLETENESS/PER-ARRAY@draft-r0.3`, owned by POINT;
- `VAL-PROFILE/POINTING-SUPPORT/POINT-DISPLACEMENT-ADMISSION@draft-r0.3`, owned by the pointing-support producer;
- `VAL-PROFILE/TELESCOPE-QC/POINT-PARAMETER-ADMISSION@draft-r0.3`, owned by the named QC process; and
- `VAL-PROFILE/CAL-TOLPROJ/POINT-AMPLITUDE-TRANSFER@draft-r0.3`, owned by CAL/TolProj.

Each draft record independently carries request, applicability, eligibility, realization, and action fields using exactly the types in Section 1.1; its current registration state is `unavailable`. Each record shall bind profile version/digest, owner, use, source facts and bindings, unavailable-state behavior, reasons, prescribed consumer action, evaluation identity, and provenance. The text of this revision neither registers a profile nor makes one evaluable. No aggregate or universal profile enters base v0.1.

3 Model, measurement, and conditional estimators

3.1 Invariant zero-background source model

For physical tangent coordinate $\boldsymbol{u}_q$, the only base model family is

$$g_q(A, \boldsymbol{\mu}, \boldsymbol{\Sigma}_g) = A \exp\left[-\tfrac{1}{2}(\boldsymbol{u}_q - \boldsymbol{\mu})^{\mathsf{T}} \boldsymbol{\Sigma}_g^{-1} (\boldsymbol{u}_q - \boldsymbol{\mu})\right], \tag{1}$$

where $\boldsymbol{\Sigma}_g$ is symmetric positive definite. The invariant principal extents are strictly positive and ordered major not less than minor. The source model has no constant, gradient, local baseline, or other nuisance term. Its six roles are amplitude, two centroid coordinates, two principal-width roles, and orientation.

The exact numerical parameter ordering, published width convention, angle origin, positive direction, gauge/period canonicalization, amplitude domain, parameter bounds, active-bound interpretation, and source-profile convention remain unavailable until `POINT-COMPATIBILITY-METHOD v0.1` is owner-approved. At exact circularity, orientation is `undefined_by_model_symmetry`; a finite placeholder is not a measurement. Exchanging principal axes with the corresponding angle transformation leaves Equation 1 invariant.

3.2 Coordinate transform and signed displacement

For parent coordinate p_q, the exact AST boundary supplies

$$\boldsymbol{u}_q = T(p_q;\ \text{coordinate state}). \tag{2}$$

The model shall use the physical tangent metric, not storage-pixel distance, unless the boundary proves their equivalence over admitted support. The ordered basis vectors, handedness, positive directions, angular unit, time and state, Jacobian where relevant, support, uncertainty, lifecycle, provenance, and failure behavior are all bound facts. Axis labels, memory order, display orientation, and FITS array order are not sign authority.

The measurement is

$$\boldsymbol{\Delta}_{\mathrm{POINT}} = \widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}_{\mathrm{expected}}, \qquad [\boldsymbol{\Delta}_{\mathrm{POINT}}] = \mathrm{arcsec}, \tag{3}$$

in that exact basis. If the tangent-coordinate origin equals the expected position, $\boldsymbol{\mu}_{\mathrm{expected}} = (0, 0)$ and $\boldsymbol{\Delta}_{\mathrm{POINT}} = \widehat{\boldsymbol{\mu}}$. The search center affects search and initialization only and never defines the zero point. POINT owns the sign in Equation 3; correction sign and composition are downstream.

3.3 Conditional numerical fit

The most that can be stated before method recovery is

$$\widehat{\boldsymbol{\theta}} = \mathrm{Resolve}_{\boldsymbol{\theta}\in\Theta_{\mathrm{effective}}} D(m, g_{\boldsymbol{\theta}}; W_{\mathrm{fit}})\,. \tag{4}$$

The admitted rows, scalar objective or likelihood D, weight meaning and normalization, constraints, comparison of multiple solutions, convergence, retry, and solution rule are not supplied. Equation 4 shall not be instantiated by guessing least squares, inverse variance, reliability weighting, or covariance weighting. Fit weights, parent stochastic covariance, inverse covariance, the formal-error model, and parent uncertainty are distinct roles.

The preserved mature procedure has configurable expected-center/central search, weighted-peak initialization, global fallback, bounded fit domain, and amplitude/width/angle constraints. Request, effective configuration, resolved plan, application attempt, and realized fit remain distinct. A sentinel shall resolve to a named effective state, and every fallback, restart, retry, branch, bound, and termination shall be reported. Exact rules belong to `POINT-COMPATIBILITY-METHOD v0.1`.

3.4 Association is not optimization

Expected position, requested search center, effective search center, selected seed, fitted centroid, and source association are distinct. Every central, global-fallback, restart, retry, or other authorized branch yields a candidate seed, never source identity. Source-attributed displacement and amplitude require `POINT-SOURCE-ASSOCIATION-STATE=established` under one branch-independent method and declared association domain. A central or global peak, finite fit, small displacement, large amplitude, dynamic range, or formal standardization cannot establish association. The numerical `POINT-FIT-AMPLITUDE-COMPONENT` may survive unavailable or failed association, but only `POINT-SOURCE-ASSOCIATED-AMPLITUDE-DIAGNOSTIC` is attributable to the intended source.

3.5 Residual and deterministic mismatch

The scientific decomposition is

$$m = g(\boldsymbol{\theta}_{\mathrm{true}}) + b + n, \tag{5}$$

where b is any deterministic parent component not represented by the zero-background model and n is stochastic variation. POINT neither fits nor silently removes b. It can bias every fit role; a finite fit does not turn it into noise. Once fitting is authorized, the fitted model and residual shall be published or reconstructible with objective, support, row count, rank, convergence, active constraints, branch state, residual diagnostics, and parent additive-reference/null state.

3.6 Conditional diagnostic identities

Only two derived product identities are admitted. The role is

`POINT-DYNAMIC-RANGE-DIAGNOSTIC/`
`FITTED-AMPLITUDE-OVER-FULL-MAP-RMS@1`.

Its conditional formula is

$$D_{\mathrm{RMS}} = \widehat{A}/\mathrm{RMS}_{\mathrm{full}}. \tag{6}$$

The role `POINT-FORMAL-AMPLITUDE-STANDARDIZATION@1` has the conditional formula

$$Z_A = \widehat{A}/\sigma_A. \tag{7}$$

Equation 6 is descriptive dynamic range only and is unavailable until `POINT-FULL-MAP-RMS-METHOD v0.1` binds the parent/generation, population, centering, weighting, source-region treatment, support/edge, non-finite and zero-denominator rules, units, lifecycle, provenance, and parent relation. It requires the exact selected fit-amplitude role, a compatible parent and generation, a finite positive denominator, and the association state required by the declared diagnostic claim. `sig2noise` may be an exact legacy alias only after `POINT-FULL-MAP-RMS-METHOD v0.1` explicitly approves that alias. `peak_over_full_map_rms` is not admitted without a separate compatibility-method finding.

Equation 7 is formal standardization only and is unavailable until `POINT-FORMAL-ERROR-METHOD v0.1` and its signed-versus-absolute convention are approved and σ_A is positive and finite. It requires the exact selected fit-amplitude role and its approved formal amplitude error. `fit_sig2noise` may be an exact legacy alias only after `POINT-FORMAL-ERROR-METHOD v0.1` explicitly approves that alias. Neither ratio is statistical significance, detection probability, false-alarm statistic, completeness, or purity. A zero or non-finite denominator produces the future method's typed unavailable or failed state, never an infinite claim.

Each derived role has its own request, parent, method, producer lifecycle, availability, support, unit-one state, cause, provenance, and claim ceiling. Neither role is required for fit-result completion. Any diagnostic described as applying to the intended source additionally requires `POINT-SOURCE-ASSOCIATION-STATE=established`; no ratio promotes an unassociated amplitude component into a source-associated role.

3.7 Conditional response objects

For one frozen branch b, resolved state g, and perturbation coordinate m, the fixed-branch parameter response, when differentiable or under another exact authorized local perturbation rule, is

$$R^{\mathrm{fixed}}_{\theta,m} = \left.\frac{\partial\widehat{\boldsymbol{\theta}}}{\partial m}\right|_{b,g,\,\mathrm{support},\,W_{\mathrm{fit}},\,\mathrm{constraints}}, \tag{8}$$

and the displacement response is

$$R^{\mathrm{fixed}}_{\Delta,m} = P_\mu R^{\mathrm{fixed}}_{\theta,m}. \tag{9}$$

For perturbation direction δm, the full-procedure response is

$$R^{\mathrm{FP}}_{\theta,m}[\delta m] = \lim_{\epsilon\to 0}\frac{\mathcal{P}(m+\epsilon\,\delta m)-\mathcal{P}(m)}{\epsilon}, \tag{10}$$

or one separately authorized finite-difference equivalent. Both invocations of $\mathcal{P}$ rerun central search, fallback, restarts and retries, support resolution, applicable weight/state resolution, constraint activation, optimization, and association. A source-domain response may be composed only with a compatible exact parent response:

$$R_{\theta,\mathrm{source}} = R_{\theta,m}R_{\mathrm{parent}}. \tag{11}$$

Parent response, fixed-branch POINT response, full-procedure POINT response, and empirical observational bias/accuracy are distinct products. Under one declared empirical population and procedure, observational displacement bias may be defined as

$$b_\Delta = \mathbb{E}\Big[\widehat{\Delta}-\Delta_{\mathrm{true}} \mid \text{population and procedure}\Big]. \tag{12}$$

No unavailable response or bias is numerically zero.

3.8 Conditional displacement covariance

When fitted-centroid and expected-position covariance objects and both cross covariances are available in one common tangent basis,

$$C_\Delta = C_{\widehat{\boldsymbol{\mu}}} + C_{\boldsymbol{\mu}_{\mathrm{expected}}} - C_{\widehat{\boldsymbol{\mu}},\boldsymbol{\mu}_{\mathrm{expected}}} - C_{\boldsymbol{\mu}_{\mathrm{expected}},\widehat{\boldsymbol{\mu}}}. \tag{13}$$

Neither cross term is zero without explicit independence authority. If a coordinate transformation is required, its Jacobian and the complete joint covariance shall be bound before transformation. If any required covariance or cross-covariance is unavailable, the combined displacement covariance is unavailable while available component uncertainties remain visible.

4 Inputs, assumptions, and dependency gates

4.1 Stage-specific lifecycle and record model

The producer lifecycle is one ordered vocabulary, not a claim that every request visits every state: `not_requested`, `requested`, `effective`, `unavailable`, `resolved`, `applied`, `fit_realized`, `complete_publication_candidate`, `publication_decided`, `published`, `failed`, `not_produced`, and `superseded`. Here `requested` is the immutable request and requested parent/method/use identities; `effective` means every sentinel and requested option has one explicit effective value or typed unavailable state; `resolved` means one exact compatible method, route, support, objective, weighting, constraints, fallback, and failure plan is selected and frozen; `applied` means that resolved procedure was invoked for one exact attempt; and `fit_realized` means immutable numerical fit values, role states, residual, and execution outcome were produced.

`complete_publication_candidate` means every required value-or-status role, identity, cause, and provenance member is structurally complete for POINT publication evaluation. `publication_decided` means the POINT-owned fit-publication/completeness decision artifact exists, and `published` means the accepted immutable candidate is exposed under its exact product role. If an execution-attempt identity named `fit_attempted` is retained in a representation, it is evidence beneath `applied`, never a competing producer-lifecycle state.

Eight logical records shall remain separate even if one representation stores them together:

A. **Request and parent record**: immutable observation, array, requested parent/method/use identities, exact parent science and provenance, source reference, requested search/fit domains, requested options, constraints, support/weight requests, and unresolved sentinels.
B. **Effective-configuration record**: every resolved sentinel and effective search, fit-domain, support, weight, constraint, and option value or typed unavailable state.
C. **Resolved-method/plan record**: one frozen method, route, support, objective, weighting, constraints, fallback, comparison, termination, and failure plan.
D. **Application-attempt record**: applied plan identity, attempt identity, invocation time/state, branch/restart/retry identity, and execution outcome.
E. **Realized-fit and component record**: selected seed, fitted values, component states, association outcome, realized constraints, residual, diagnostics, response/bias/uncertainty states, causes, and provenance.
F. **Publication-candidate record**: every required value-or-status role with identity, cause, provenance, and structural-completeness state.
G. **Publication decision**: POINT-owned fit-completeness profile, decision, reasons, decision identity, lifecycle, and provenance.
H. **Published product record**: exact accepted immutable candidate, product role, version, generation, exposure identity, lifecycle, and provenance.

The request and parent record shall not require a selected seed, fitted centroid or parameters, association outcome, realized constraint or residual, or applied/fit-realized outcome. Those facts belong only to later records.

Every route boundary instance shall additionally bind its exact predecessor package/version/source digest, signal role, owner approval, compatibility/supersession status, source morphology, response chain, processed-map profile, Gaussian compatibility, response-center relation, model-mismatch state, expected centroid bias or typed unavailability, fit-domain relation, and displacement-claim status.

4.2 Method authorities and present availability

The base operator identity is `POINT-FIT/ELLIPTICAL-GAUSSIAN-COMPATIBILITY@1`. Its numerical realization is gated by three scientifically and lifecycle-distinct records:

Authority	Current state	Consequence
`POINT-COMPATIBILITY-METHOD v0.1`	pending separate owner approval	No numerical fit or fit-derived product. The future record must bind model parameterization, rows, objective, weights, support, search, initialization, association, fallback, constraints, solver-independent resolution, termination, rank, identifiability, sentinel, failure, identity, and compatibility status.

Authority	Current state	Consequence
`POINT-FORMAL-ERROR-METHOD v0.1`	pending separate owner approval	Marginal formal errors, formal standardization, and formal-uncertainty-dependent uses are unavailable. An otherwise authorized fit is not erased. Joint covariance remains unavailable unless separately authorized.
`POINT-FULL-MAP-RMS-METHOD v0.1`	pending separate owner approval	Only `fitted_amplitude_over_full_map_rms` and exact legacy alias `sig2noise` are blocked. An otherwise authorized fit, displacement, effective shape, or formal error is not erased.

The exact numerical parent-route instances, registered named-use profiles, fixed-branch response, full-procedure response, and observational bias/accuracy are also unavailable. No familiar practice, optimizer output, method name, finite value, or adjacent-package analogy may fill any gap.

4.3 Product and claim dependency matrix

Product or claim	Required available facts	Blocking result
Fit result	Exact numerical parent and tangent metric; approved compatibility method; admitted support; objective/weight state; successful realization; component states	No numerical fit or fit-derived product.
Fit components	Conforming fit and exact component-identifiability state	Only the affected component is unavailable or failed; conforming siblings remain.
Source association	Exact source-reference authority and branch-independent method/domain	Source attribution is unavailable or failed.
Processed-map displacement	Fitted centroid; expected position; identical tangent chart/basis; established association	No source-attributed displacement. Response and centroid bias may be typed unavailable but remain visible.
Fit-amplitude component	Conforming fitted A role, exact parent unit/calibration/response, support, constraint, and component state	Component unavailable or failed; no amplitude-derived role.
Source-associated amplitude	Fit-amplitude component plus established association	No source-attributed amplitude; numerical fit component remains visible.
Unbiased pointing or correction-use claim	Displacement plus every named-use-required response center, centroid bias, uncertainty, and pointing-support policy	Named-use eligibility is `ineligible` or `decision_unavailable`; missing bias is not zero.
Formal errors	Conforming fit and approved formal-error method	Formal errors unavailable; fitted values may remain.
Formal standardization	Exact selected fit-amplitude role; approved formal-error method; finite positive formal amplitude error; approved sign convention	Independently requested diagnostic unavailable.
Dynamic-range diagnostic	Exact selected fit-amplitude role; approved full-map-RMS method; compatible parent/generation; finite positive denominator; claim-appropriate association	Independently requested diagnostic unavailable.
Photometric transfer	Source-associated amplitude plus CAL/TolProj profile and source/reference authority	Photometric eligibility is `ineligible` or `decision_unavailable`.
Telescope-QC display	Required fit facts plus QC-owner applicability, eligibility, realization, and action	Realization is `not_produced`, `incomplete`, or `failed`; it cannot rescue another use.

Response and covariance availability are evaluated by the applicable row and named-use policy. Missing response or covariance neither universally erases all processed-map results nor authorizes response, bias, uncertainty, correction, calibration, or photometric claims.

4.4 Uncertainty budget

Conditional formal fit error, joint parameter covariance, source-reference or ephemeris uncertainty, WCS/tangent-coordinate uncertainty, parent response/model-mismatch uncertainty, calibration uncertainty, empirical pointing repeatability, NOI uncertainty, cross-array dependence, and pointing-correction uncertainty are separate components with separate owners. They may be combined only when every variance and cross-covariance needed for the declared operation is available. Missing covariance is not zero, diagonal covariance, or independence.

When the fitted-centroid and expected-position covariances are expressed in one common tangent basis, Equation 13 governs their conditional combination. Neither fitted/expected cross-covariance term may be dropped without independence authority. Coordinate transformations require their exact Jacobians and complete joint covariance before transformation. Fit-conditional centroid covariance, source-reference or ephemeris covariance, AST coordinate uncertainty, parent response/model mismatch, empirical repeatability, observational centroid bias, cross-array dependence, and correction uncertainty remain separate.

If fit and published tangent bases differ, exact transformation generally requires joint centroid covariance. Transforming marginal errors alone through a non-axis-aligned Jacobian does not produce exact AltAz marginal errors. A later joint-covariance, astrometric, empirical-repeatability, or NOI companion is separately versioned and does not rewrite the immutable POINT fit claim.

4.5 Response and bias separation

POINT can be nonlinear because search, fallback, support, or active constraints can depend on the data. A fixed-branch response is conditional on one exact seed/fallback branch, fit support, weight state, and active-constraint set. A full-procedure response reruns every data-dependent search, fallback, support, constraint, and optimization choice under perturbation. Observational centroid bias and pointing accuracy require a separate empirical authority. Source association is separate from all three. Parent response is input state, not automatically fitter response; a local Jacobian is not full-procedure response. Equations 8–12 define the domains and codomains without authorizing a construction. All three POINT response/bias roles are presently unavailable under separate release conditions, and their absence does not imply zero response error or zero centroid bias.

4.6 Named-use ownership

POINT owns only the per-array fit-completeness policy. The pointing-support producer owns displacement admission. The named telescope/observing QC process owns parameter admission and actions. CAL/TolProj owns amplitude admission for photometric transfer. The same immutable result may pass one use, fail another, and be shown diagnostically for a third. No state or action propagates across uses. Failure of one array does not erase siblings; POINT imputes no array and publishes no observation-wide success. A downstream partial-set aggregate must record its exact policy and membership.

4.7 Claim classes and evidence ceiling

This contract establishes scientific meaning and conditional engineering conformance requirements. It does not establish representation fidelity, implementation conformity, numerical production, response or covariance fidelity, uncertainty coverage, observational validation, pointing accuracy, performance, readiness, production suitability, production authorization, or Unity activity.

5 Validity, unavailable states, and edge cases

5.1 Typed unavailable-state register

ID	Unavailable quantity or claim	Release condition or disposition
SCI-POINT-UNAV-001	POINT-owned cross-array aggregate	Outside base v0.1; successor decision required.
SCI-POINT-UNAV-002	POINT correction candidate or applied correction	Outside scope; pointing-support producer and AST retain ownership.
SCI-POINT-UNAV-003	Numerical MAP/JINC/FLT route	Exact predecessor authority, product, state, and compatibility binding.
SCI-POINT-UNAV-004	Numerical compatibility fit and every fit-derived product	Approved compatibility method, exact numerical parent, and route binding.

ID	Unavailable quantity or claim	Release condition or disposition
SCI-POINT-UNAV-005	POINT whole-observation success from partial arrays	Outside scope; downstream owns partial-set admission.
SCI-POINT-UNAV-006	Full joint covariance or uncertainty coverage	Separately authorized representation and evidence.
SCI-POINT-UNAV-007	Absolute flux or calibration accuracy	CAL/TolProj authority and evidence; never POINT alone.
SCI-POINT-UNAV-008	Intrinsic beam inferred from Pointing width	Separate authorized method; SCI-BEAM boundary.
SCI-POINT-UNAV-009	Statistical significance or detection probability	Complete probabilistic method and validation.
SCI-POINT-UNAV-010	Applied correction for another observation	Producer selection and AST application authority.
SCI-POINT-UNAV-011	OOF or blank-field source result	Separate package authority.
SCI-POINT-UNAV-012	Coadd or intermediate-FRUIT parent	Outside base v0.1.
SCI-POINT-UNAV-013	Universal cross-use eligibility or aggregate profile	Prohibited; policies remain use-specific.
SCI-POINT-UNAV-014	Source-attributed displacement or amplitude	Valid source reference and established association.
SCI-POINT-UNAV-015	Published AltAz displacement	Exact AST/source-to-POINT basis boundary and valid transform.
SCI-POINT-UNAV-016	Marginal formal errors and dependent uses	Approved formal-error method; fit values may remain.
SCI-POINT-UNAV-017	$\mathrm{RMS}_{\mathrm{full}}$ scale product	Approved full-map-RMS method binding exact population, centering, weighting, support, source-region, edge, finite, unit, generation, and provenance rules.
SCI-POINT-UNAV-018	Fixed-branch POINT response	Approved compatibility method plus exact local perturbation construction with branch, support, weights, constraints, and state frozen.
SCI-POINT-UNAV-019	Derived fitted-amplitude/full-map-RMS diagnostic and `sig2noise` alias	UNAV-017 released, compatible selected fit-amplitude role and parent generation, positive finite denominator, claim-appropriate association, and explicit alias approval.
SCI-POINT-UNAV-020	`known/isolated/bright/approximately_centered` fact	Exact fact authority or test and cause.
SCI-POINT-UNAV-021	Exact numerical parent boundary instance	Exact package/version/source digest, role, owner approval, compatibility, and boundary state.
SCI-POINT-UNAV-022	Full-procedure POINT response	Separate complete-rerun authority covering every data-dependent branch under perturbation.
SCI-POINT-UNAV-023	Observational bias/accuracy	Separate empirical population, procedure, reference-truth, and evidence authority.

5.2 Edge-case rules

Condition	Required behavior
Missing method, boundary, input, or applicability authority	The dependent quantity is `unavailable` with identity and cause; no default is inferred.
Attempted computation cannot conform	Report `failed`; do not relabel it scientifically unavailable.
Missing or non-finite admitted support	Apply the future compatibility method's exact failure/unavailability rule; never silently delete rows.
Active parameter bound	Report the constraint and `available_bound_censored` where applicable.

Condition	Required behavior
Weak curvature or near degeneracy	Report `weakly_identified`; formal error may be unavailable independently of the fitted value.
Exact circularity	Orientation is `undefined_by_model_symmetry`.
Major/minor exchange	Enforce canonical ordering and transform orientation consistently without changing the invariant model.
Centroid on a search or fit boundary	Report boundary/constraint state; named-use eligibility is evaluated separately.
Rank-deficient or non-positive curvature	Formal error is unavailable unless the future exact method defines another conforming result; fit-component state is assessed separately.
Unmodeled constant, gradient, or structure	Preserve residual and parent additive-reference state; do not fit or silently subtract it.
Association unavailable or failed	A numerical fit may remain a diagnostic, but source-attributed amplitude and displacement are unavailable.
Missing profile or source binding	Apply Section 1.2: applicability is `applicability_unknown`, eligibility is `decision_unavailable`, and realization is `not_produced`; preserve producer facts.
Structural named-use conflict	Apply Section 1.2: applicability is `applicability_unknown`, eligibility is `decision_unavailable`, and realization is `failed`.
Named use not requested or inapplicable	Emit the exact request/applicability/eligibility/realization tuple from Section 1.2; never substitute producer `unavailable`.
Incomplete, failed, or absent requested artifact	Use realization `incomplete`, `failed`, or `not_produced`, respectively; eligibility remains separately typed.
Zero or non-finite diagnostic denominator	Ratio is unavailable or failed under its future exact method, never infinite.
Unavailable response or covariance	Follow the dependency matrix; do not universally erase or authorize claims.
One array unavailable or failed	Preserve sibling atomic products; publish no POINT observation-wide success.
Terminal-FRUIT assertion absent, or intermediate iteration	Reject the parent for base v0.1.
Requested route unavailable	Do not choose, substitute, or fall back to another parent route.
Diagnostic-display action present	Preserve the four named-use fields; the action cannot rescue any other exclusion.
One fit component unavailable or failed	Preserve every conforming sibling role and the complete value-or-status atom.

6 Engineering record contracts

The representation syntax is implementation-defined, but the following logical records and fields shall be lossless, queryable, and immutable for a published result. Co-location in one table or file does not merge identities.

6.1 Stage-specific record separation

Record	Required logical fields
A. Request and parent	Immutable request; observation/array; requested parent, method, and uses; exact parent identity/science; source reference; `known`, `isolated`, `bright`, and `approximately_centered` facts; requested search/fit domains, initialization, fallback, constraints, support/weights, and unresolved sentinels. No selected seed, fit value, association outcome, realized constraint/residual, or applied/realized outcome.
B. Effective configuration	Every sentinel and requested option resolved to one explicit effective value or typed unavailable state.

Record	Required logical fields
C. Resolved method/plan	One frozen compatible method, route, support, objective, weighting, constraints, fallback, comparison, termination, and failure plan.
D. Application attempt	Applied plan, invocation/attempt identity, branch/restart/retry identity, execution state, cause, and provenance.
E. Realized fit/components	Selected seed; numerical values; component states; association outcome; realized constraints; residual; diagnostics; response, bias, and uncertainty states; causes and provenance.
F. Publication candidate	Every required value-or-status role, identity, cause, provenance, and structural-completeness state.
G. Publication decision	POINT fit-completeness profile and source bindings; decision, reasons, identity, lifecycle, and provenance.
H. Published product	Accepted immutable candidate; exact role, version, generation, exposure identity, lifecycle, and provenance.

6.2 Realized fit and component record

When a numerical fit is authorized, record E shall carry branch and selected seed or peak identity; association method, domain, state, and cause; admitted support and row count; exact objective and fit-weight identity; parameter values with units and conventions; per-component identifiability; active constraints; rank; termination, convergence, and retry; fitted model or reconstructibility; residual and diagnostics; parent additive-reference and null state; fixed- and full-procedure response states; observational bias and accuracy state; formal-error and covariance identities and states; and every typed cause of unavailability or failure. Its numerical A role is

`POINT-FIT-AMPLITUDE-COMPONENT`.

Only established association may create the separately typed source-associated amplitude role:

`POINT-SOURCE-ASSOCIATED-`
`AMPLITUDE-DIAGNOSTIC`.

The source association record shall cover every central, global-fallback, restart, retry, or other branch. It is a separate product role and cannot be derived from fit quality.

6.3 Displacement record

A displacement record shall contain the associated fit identity; expected source-position identity; fitted centroid; exact ordered basis vectors; handedness and positive directions; arcsecond unit; coordinate state/time; parent-to-tangent transform and Jacobian where relevant; source and coordinate uncertainty states; Equation 3 sign declaration; support, validity, lifecycle, and provenance. It shall contain no implied correction sign or applied-correction claim.

6.4 Named-use evaluation record

For each named use, record profile identity/version/digest, policy owner, source bindings, request, applicability, eligibility, realization, reason, prescribed consumer action, evaluation identity, lifecycle, and provenance. The four axes shall be the exact independent SCI-VAL enumerations: request is `requested` or `not_requested`; applicability is `applicable`, `inapplicable`, or `applicability_unknown`; eligibility is `eligible`, `ineligible`, or `decision_unavailable`; realization is `realized`, `incomplete`, `failed`, or `not_produced`. The evaluator shall apply Section 1.2. The action field may contain `diagnostic_display_only`; that token is no axis or producer state. No result from one named use may modify another.

6.5 Availability precedence

For each product request, evaluate in order: exact parent and route boundary; method authority; coordinate/source boundary; applicability facts; attempted producer application and realization; component identifiability; source association; the product row in Section 4.3; then each requested named-use profile. Stop only the dependent product or claim. Preserve independent siblings and record the exact blocking state.

6.6 Response and covariance records

A response record shall identify its domain, codomain, perturbation, selected fit roles, branch policy, support, weights, constraints, method, finite versus differential construction, parent response compatibility, state, cause, and provenance. It shall identify exactly one of fixed-branch response, full-procedure response, parent response, or empirical bias/accuracy. A conditional displacement-covariance record shall bind every term of Equation 13, the common basis, transformations and Jacobians, independence authority where any cross term is zero, validity, state, cause, and provenance.

7 Method-record acceptance criteria

`POINT-COMPATIBILITY-METHOD v0.1` shall not become available unless one immutable, content-bound, owner-approved record specifies the exact source equation and parameter ordering; width and angle conventions; domains and constraints; rows and residual; objective/likelihood; weight meaning and covariance relation; support; expected-position/search-center distinction; central and global search; score, candidate, interpolation, tie, initialization, fallback, association, retry, and sentinel rules; solver-independent solution comparison; convergence, rank, identifiability, failure, and compatibility status; and numerical tests for each applicability class or typed unavailability.

`POINT-FORMAL-ERROR-METHOD v0.1` shall separately specify the fitted parameterization, stochastic model, weight relation, exact covariance/error construction, observed/expected curvature, residual scaling, degrees of freedom, rank and positive-definiteness rules, transformations, marginal extraction, active bounds, circularity/axis degeneracy, weak identification, consistency checks, units, scope, omitted uncertainty, and unavailable/failed states.

`POINT-FULL-MAP-RMS-METHOD v0.1` shall separately specify the exact signal parent and generation, full-map population, about-zero RMS versus centered deviation, center/mean, weighting, source-region treatment, support/edge/validity, non-finite and zero-denominator behavior, units, lifecycle/provenance, and identity relation to the amplitude parent.

The method records shall also state whether and how `sig2noise` or `fit_sig2noise` is an exact legacy alias. No alias is admitted merely by this contract text, and `peak_over_full_map_rms` requires a separate compatibility-method finding.

8 Normative conformance requirements

An implementation conforms to one requirement only when the exact records and states named by that requirement are present and valid. A finite numerical value does not substitute for a missing identity, authority, or state.

Requirement	Normative statement	Scientific source
SCI-POINT-REQ-001	Treat every requested observation-array-parent-method fit as an independent value-or-status scientific atom and keep its request, effective, resolved, applied, realized, candidate, decision, and published records distinct.	Secs. 2, 4.1
SCI-POINT-REQ-002	Bind every stage-specific record field in Section 4.1; do not infer identity from filenames, labels, location, or facts belonging to a later stage.	Sec. 4.1
SCI-POINT-REQ-003	Admit only exact MAP, JINC, FLT-FIXED, or FLT-MATCHED signal roles and preserve their distinct meanings and companions.	Sec. 2.2
SCI-POINT-REQ-004	Never automatically select, substitute, equate, or fall back among parent routes.	Sec. 2.2
SCI-POINT-REQ-005	Treat FRUIT only as complete terminal ancestry on one admitted route; reject intermediate iterations and coadds.	Sec. 2.2
SCI-POINT-REQ-006	Bind the exact route response chain, processed profile, Gaussian compatibility, response center, model mismatch, centroid-bias state, and claim status before numerical use.	Secs. 4.1, 2.2
SCI-POINT-REQ-007	Bind expected source position, parent WCS origin, search center, and fitted centroid as four distinct objects.	Secs. 4.1, 3.2
SCI-POINT-REQ-008	Use the exact ordered AltAz tangent basis, physical metric, transform, handedness, positive directions, state, support, uncertainty, and provenance.	Eqs. 2–3
SCI-POINT-REQ-009	Publish displacement only as fitted centroid minus expected position in arcseconds; do not encode correction sign.	Eq. 3
SCI-POINT-REQ-010	Use only the zero-background six-role elliptical-Gaussian family in Equation 1; add no background or second model family.	Sec. 3.1

Requirement	Normative statement	Scientific source
SCI-POINT-REQ-011	Preserve the invariant positive-definite shape, canonical principal ordering, and symmetry-defined orientation state without inventing numerical width or angle conventions.	Sec. 3.1
SCI-POINT-REQ-012	Produce no numerical fit or fit-derived product until the approved compatibility method, exact parent, and route binding are available.	Sec. 4.2
SCI-POINT-REQ-013	Keep objective, weights, stochastic covariance, inverse covariance, formal-error model, and parent uncertainty distinct; never guess Equation 4.	Sec. 3.3
SCI-POINT-REQ-014	Use the exact producer lifecycle in Section 4.1; distinguish the frozen resolved plan from its application attempt and realized fit, and preserve every search, support, fallback, retry, constraint, sentinel, and termination state.	Secs. 4.1, 3.3
SCI-POINT-REQ-015	Establish source association independently for every authorized search, fallback, restart, and retry branch before creating any source-associated displacement, amplitude, or intended-source diagnostic.	Sec. 3.4
SCI-POINT-REQ-016	Carry `known`, `isolated`, `bright`, and `approximately_centered` applicability as typed facts with authority, domain, cause, validity, lifecycle, and provenance.	Sec. 2.4
SCI-POINT-REQ-017	When fitting is authorized, place selected seed, fitted values, association outcome, realized constraints, model/residual, objective, support, row count, rank, convergence, branch, diagnostics, and additive-reference/null state only in the applicable post-request records and publish or make them reconstructible.	Secs. 4.1, 3.5
SCI-POINT-REQ-018	Give every required fit component exactly one component-identifiability state and preserve available sibling components when another is unavailable or failed.	Secs. 1.1, 4.3
SCI-POINT-REQ-019	Type the numerical A role as `POINT-FIT-AMPLITUDE-COMPONENT`; create `POINT-SOURCE-ASSOCIATED-AMPLITUDE-DIAGNOSTIC` only after established association, with exact parent unit, calibration, response, constraint, support, and uncertainty context.	Secs. 2.3, 3.4
SCI-POINT-REQ-020	Publish widths and angle only as processed-map effective-shape and QC roles; do not claim intrinsic beam or unique cause.	Secs. 2.3, 3.1
SCI-POINT-REQ-021	Publish marginal formal errors only under the separately approved formal-error method with assumptions, conditioning, domain, units, and limitations.	Secs. 4.2, 4.4
SCI-POINT-REQ-022	Represent absent joint covariance as unavailable, not zero, diagonal, or independence, and preserve later uncertainty companions as separate versions.	Sec. 4.4
SCI-POINT-REQ-023	Treat the two roles in Equations 6–7 as independently requested derived products with exact amplitude, method, parent, lifecycle, availability, support, unit-one, cause, provenance, association, alias, and claim-ceiling gates.	Sec. 3.6
SCI-POINT-REQ-024	Do not admit `peak_over_full_map_rms` as an alias unless a future compatibility-method finding separately authorizes it.	Sec. 3.6
SCI-POINT-REQ-025	Keep fixed-branch response, full-procedure response, observational bias/accuracy, source association, and parent response distinct.	Sec. 4.5
SCI-POINT-REQ-026	Apply the product-and-claim dependency matrix by exact role, including fit-amplitude versus source-associated amplitude and both derived diagnostics; missing association, response, covariance, or method shall neither erase nor authorize unrelated facts.	Sec. 4.3
SCI-POINT-REQ-027	Keep the exact producer lifecycle, component identifiability, and SCI-VAL request/applicability/eligibility/realization types independent and lossless; never use one axis's token on another.	Secs. 1.1, 1.2
SCI-POINT-REQ-028	Use `diagnostic_display_only` only as a post-evaluation consumer action; never use it or `diagnostic_only` as an axis value, producer lifecycle state, or universal flag.	Secs. 1.1, 1.2
SCI-POINT-REQ-029	Preserve separate policy ownership for fit completeness, displacement admission, telescope-QC admission, and source-associated photometric-transfer admission; SCI-VAL only registers and evaluates exact profiles.	Secs. 2.1, 2.5

Requirement	Normative statement	Scientific source
SCI-POINT-REQ-030	Treat all four named-use profile identities and their exact four-axis mechanics as unapproved drafts pending final owner review; this textual revision neither registers nor makes them evaluable, and no aggregate/universal profile may be constructed.	Secs. 2.5, 1.2
SCI-POINT-REQ-031	Publish no POINT cross-array aggregate, correction, correction selection, correction application, or whole-observation success.	Secs. 2, 4.6
SCI-POINT-REQ-032	Do not impute a missing array or erase conforming siblings after one array's failure.	Secs. 2, 4.6
SCI-POINT-REQ-033	Keep deterministic unmodeled background/structure visible as residual and parent state; do not fit or silently subtract it.	Eq. 5
SCI-POINT-REQ-034	Apply every typed unavailable state, SCI-VAL disposition, and edge rule in Section 5, including exact cause, release condition, and preservation of conforming sibling facts.	Secs. 5, 1.2
SCI-POINT-REQ-035	Keep all uncertainty-budget components separate and form conditional displacement covariance only by Equation 13 with every required variance, cross-covariance, basis, and Jacobian available.	Secs. 4.4, 3.8
SCI-POINT-REQ-036	Publish a full-procedure response only when every data-dependent branch is rerun under perturbation; do not infer empirical accuracy from either response role.	Sec. 4.5
SCI-POINT-REQ-037	Evaluate every named use with the exact SCI-VAL disposition function and four enumerations before any action, including missing-profile, missing-source, conflict, not-requested, inapplicable, incomplete, failed, and not-produced cases, without cross-use rescue or propagation.	Secs. 1.2, 4.6
SCI-POINT-REQ-038	Respect the evidence ceiling: this contract alone authorizes no implementation, numerical, fidelity, validation, accuracy, performance, readiness, production, or Unity claim.	Sec. 4.7

9 Falsifiable predictions and traceability

These predictions specify tests that a future implementation or representation can fail. They are not claims that such validation has occurred.

Prediction	Falsifiable consequence	Governing requirements
SCI-POINT-PRED-001	For an exact noiseless matching Gaussian with complete admitted support, identifiable noncircular shape, parameters in the admitted interior or under the approved boundary rule, and one unique compatibility-method solution, the method recovers canonical amplitude, centroid, and invariant shape. At circularity the equal widths and invariant shape recover while orientation is symmetry-undefined; at an active bound the affected role is bound-censored under the approved constrained solution.	REQ-010,011,012,013,014,018
SCI-POINT-PRED-002	Authorized scaling of the parent signal and compatible unit scales amplitude as the method declares without moving the source reference.	REQ-006,007,019
SCI-POINT-PRED-003	Positive displacement follows fitted centroid minus expected position; correction sign remains downstream.	REQ-008,009,031
SCI-POINT-PRED-004	Changing only the search center may change branch or seed but never measurement zero.	REQ-007,009,014
SCI-POINT-PRED-005	WCS handedness or basis changes transform components through the exact boundary, independent of memory/display order.	REQ-008,009
SCI-POINT-PRED-006	Central-search and global-fallback branches remain distinguishable across resolved plan, applied attempt, and realized-fit records.	REQ-001,014
SCI-POINT-PRED-007	An exact score tie follows the future method's deterministic tie rule and records the selected result only after application, never in the immutable request.	REQ-002,012,014
SCI-POINT-PRED-008	A wrong global peak may retain a fit-amplitude component but cannot yield source-associated displacement, amplitude, or intended-source diagnostic without established association.	REQ-015,019
SCI-POINT-PRED-009	Principal-axis exchange with corresponding angle transformation leaves the Gaussian invariant.	REQ-011
SCI-POINT-PRED-010	Exact circularity makes orientation undefined by model symmetry.	REQ-011,018
SCI-POINT-PRED-011	Active amplitude, width, or angle constraints appear in applied and realized records; affected components are bound-censored where applicable and are not silently counted as unconstrained recovery.	REQ-014,018
SCI-POINT-PRED-012	An unmodeled constant or gradient can bias fit roles and remains visible in residual/additive-reference state.	REQ-017,033
SCI-POINT-PRED-013	Missing or non-finite support follows the exact future method rule, is never silently deleted, and leaves request, effective, resolved, applied, and realized evidence in their proper records.	REQ-002,014,034
SCI-POINT-PRED-014	Reliability weights cannot be interpreted as inverse covariance without explicit authority.	REQ-013

Prediction	Falsifiable consequence	Governing requirements
SCI-POINT-PRED-015	Rank deficiency, active bounds, non-positive curvature, or angle degeneracy can preserve fitted values while formal errors remain unavailable.	REQ-018,021,034
SCI-POINT-PRED-016	MAP/JINC/FLT response mismatch changes compatibility or bias state and is not erased by common Gaussian naming.	REQ-003,006,025
SCI-POINT-PRED-017	Parent-response asymmetry may yield typed centroid-bias state rather than assumed zero bias.	REQ-006,025
SCI-POINT-PRED-018	A terminal FRUIT product retains its exact terminal route and ancestry; an intermediate iteration is rejected.	REQ-005
SCI-POINT-PRED-019	Failure of one array or one component preserves conforming sibling atoms and roles and produces no POINT observation-wide success.	REQ-001,018,031,032
SCI-POINT-PRED-020	No input produces a POINT-owned cross-array aggregate or telescope correction.	REQ-031
SCI-POINT-PRED-021	The independently requested dynamic-range role follows the exact RMS and selected-amplitude authorities and is unavailable for a zero, non-finite, or unavailable denominator or missing claim-required association.	REQ-019,023,034
SCI-POINT-PRED-022	The independently requested formal-standardization role is unavailable without its exact selected amplitude, positive finite approved formal error, and sign convention, and never becomes statistical significance by naming.	REQ-019,021,023
SCI-POINT-PRED-023	One fit may have different exact SCI-VAL tuples across uses; `diagnostic_display_only` changes no producer lifecycle, axis value, or sibling use.	REQ-027,028,037
SCI-POINT-PRED-024	Missing profile or required source binding yields `applicability_unknown`, `decision_unavailable`, and `not_produced`; a structural conflict yields `failed`; none erases available producer facts.	REQ-027,030,034,037
SCI-POINT-PRED-025	Fixed-branch response cannot be reported as full-procedure response, and neither establishes empirical accuracy.	REQ-025,036
SCI-POINT-PRED-026	Missing compatibility method yields no numerical fit or fit-derived product.	REQ-012
SCI-POINT-PRED-027	With compatibility fitting available but formal-error authority missing, fitted values may exist while formal errors, unsupported joint covariance, and formal standardization remain unavailable.	REQ-021,022,026
SCI-POINT-PRED-028	Missing full-map-RMS authority blocks the RMS scale product, independently requested dynamic-range role, and any explicitly approved alias, not an otherwise authorized fit, fit-amplitude component, displacement, or formal error.	REQ-019,023,026
SCI-POINT-PRED-029	Every search, fallback, restart, and retry branch requires exact association before source-associated amplitude, displacement, or any diagnostic claimed for the intended source.	REQ-015,019,023
SCI-POINT-PRED-030	Fixed-branch response, full-procedure response, observational bias/accuracy, or covariance unavailability follows its separate release condition and neither universally erases nor authorizes other roles and claims.	REQ-025,026,035,036
SCI-POINT-PRED-031	A `diagnostic_display_only` action cannot rescue correction, photometric, fit-completeness, realization, or another named-use exclusion.	REQ-028,037
SCI-POINT-PRED-032	A shared expected-source reference error contributes through the covariance and cross-covariance terms of Equation 13; it does not average down as independent per-array noise without explicit independence authority.	REQ-035

10 Owner-decision parity and remaining approvals

The engineering contract has one-to-one coverage of ODQ-001 through ODQ-009, including ODQ-003A/003B, and preserves the three distinct method dispositions. The generic conditional science contains no unresolved ODQ. Remaining owner actions are: content-bound approval of the three method records; approval of exact numerical route instances; final review and registration of the four draft profile identities/mechanics; and separate authorization of response or observational bias/accuracy authorities if those claims are sought.

The r0.2 owner/view parity report records gap-free REQ-001–REQ-038 and PRED-001–PRED-032, exact common-core imports in both views, canonical token checks, and the distinct release conditions for fixed-branch response, full-procedure response, and observational bias/accuracy. The complete delivery packet contains the exact Stage A archive and sidecars, common core, both document sources, boundary/profile/method templates, traceability, manifests, build recipe, and QA records.

11 Nonclaims

This specification records no implementation inspection and claims no implementation conformity, numerical production, representation fidelity, response/covariance fidelity, uncertainty coverage, observational validation, pointing accuracy, performance, readiness, production suitability, production authorization, or Unity activity.